

Text Simplification for Children: Evaluating LLMs vis-à-vis Human Experts

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Abstract

Large Language Models (LLMs) can facilitate teaching and learning by generating educational content for specific grade levels. Generation of age and grade-appropriate materials often involves text simplification. Previous work evaluating LLMs' text simplification for children in educational context showed that LLMs can reduce lexical complexity of a text and improve its readability, but they often use vocabulary that is still too difficult for the targeted grade. In this study we focus on GPT-4o, the most advanced LLM at the time of writing. We evaluate its ability to simplify text for elementary school children and compare its performance vis-à-vis the human baseline. We show that GPT-4o can successfully reduce text complexity on lexical, syntactic, and discourse levels. However, compared to human experts, it generates syntactically more complex text. This suggests that further fine-tuning with focus on syntax is needed before LLM-generated output reaches the intended user group.

CCS Concepts

- **Human-centered computing** → **Empirical studies in HCI**;
- **Computing methodologies** → *Artificial intelligence*; Natural language processing; Natural language generation.

Keywords

Text Simplification, Children, LLMs, GPT-4o, Syntactic Complexity

ACM Reference Format:

Anastasia Smirnova, Kyu beom Chun, Wil Louis Rothman, and Siyona Sarma. 2025. Text Simplification for Children: Evaluating LLMs vis-à-vis Human Experts. In *Extended Abstracts of the CHI Conference on Human Factors in*

Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3706599.3719889>

1 Introduction

Rapid development of Large Language Models (LLMs), technology that exhibits sophisticated human-like linguistic behavior, has a great potential to change how humans access information and learn. LLMs' emergent abilities, i.e. abilities that are not attested in smaller-scale models [50], allow LLMs to perform well on tasks that previously required human expertise and skill. One example of LLMs' potential applications is for generating informational and educational content for specific demographic groups.

Even in Western democracies, information in the public domain is not accessible to all. Plain language and easy-to-read language are specifically designed to facilitate information access for non-proficient adults (adults with low literacy levels, non-native speakers) and cognitively divergent individuals, respectively [11, 16, 35]. Such content is expensive to produce, and it covers only a fraction of publicly available information. LLMs can assist experts with generation of plain language and easy-to-read materials and fine-tune them for diverse user groups [15, 27]. Similarly, in education, production of age-appropriate materials that match developmental and reading levels of users is a laborious, time-consuming process traditionally performed by human experts. LLMs can facilitate learning and teaching processes by generating age and grade-appropriate educational content [1, 8, 22, 34, 54]. Moreover, in developing countries where population literacy levels are low, the integration of LLMs in educational practices can remove barriers for participation and make education more inclusive [44].

Content generation for specific groups of users usually involves text simplification, a process that reduces lexical and structural complexity of the original text, while preserving its meaning [46, 47]. Text simplification with LLMs has received a great deal of attention in the CHI community, with the applications ranging from facilitating language acquisition for L2 learners [24] and producing materials for people with intellectual disabilities [43] to task guidance in augmented reality [52]. The current paper contributes to the emergent field of research on leveraging LLMs' capabilities to generate user-appropriate content by means of text simplification.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719889>

We specifically focus on evaluating GPT-4o’s ability to generate grade-appropriate educational content for elementary school children.

Recent studies on the topic returned mixed results. [33] investigated how well ChatGPT can support information discovery by 4th graders in the context of a history curriculum. The authors evaluated linguistic and stylistic complexity of ChatGPT-generated content in response to two different types of prompts: general prompts and child-specific prompts. The authors found that while child-specific prompts returned linguistically and stylistically simpler answers compared to the general prompts, the readability level of GPT-generated content for children was still much higher than the intended grade level. Similarly, [48] investigated LLMs’ ability to generate and simplify stories for young children with the special attention to age-appropriate vocabulary. They found that while LLMs were generally able to use simple, age-appropriate words, they also tended to include vocabulary for older age groups. LLM-generated stories also had worse readability score compared to that for human-written stories. In a related line of research, [39] investigated LLMs’ abilities to simplify text of math problems for elementary school children. The analysis showed that GPT-simplified texts showed improvement in readability metrics compared to the original texts. However, the accuracy of text simplifications was not particularly high. The authors concluded that LLM-generated simplifications can be used as suggestions but not as a stand-alone application.

Our contribution differs from the previous work in several respects. First, we use the latest and most advanced LLM at the time of writing, GPT-4o by OpenAI, which provides an up-to-date view on the performance of this technology. Second, we analyze GPT-4o’s performance with respect to two text types, original texts and human-simplified texts produced by experts. We argue that the evaluation of LLMs with respect to original, non-simplified texts allows for a better understanding of text simplification as an emergent ability in LLMs. At the same time, the comparison of LLMs-simplified texts and human-simplified texts provides a more nuanced view on how LLMs fare with respect to human experts. Finally, we evaluate the performance of GPT-4o on a comprehensive set of text complexity metrics, encompassing lexical, syntactic, and discourse cohesion features, thus extending the scope of evaluation reported in previous studies, which mainly focus on lexical features.

Our main finding is that GPT-4o performs well on text simplification tasks, overcoming difficulties reported for prior models. Our work also points to an important difference between GPT-4o and humans in that texts produced by the former are syntactically more complex.

2 Text Simplification and Measures of Text Complexity

At the level of text, complexity can be measured by different metrics, each targeting a specific property of text, ranging from surface features such as word and sentence length to organization of information in discourse [3, 37, 40, 47]. At the level of the user, the ultimate beneficiary of text simplification, complexity can be measured in terms of readability and comprehension. Thus, different

complexity-reducing operations applied to texts, such as replacement of difficult words with more familiar synonyms or splitting long sentences into several shorter ones, aim to facilitate cognitive processes that support reading and comprehension. These processes include word identification and recognition, sentence representation, and construction of meaning [29, 49].

Before simplified texts are presented to the target demographic groups for evaluation, the analysis of complexity at the textual level is a necessary step [45]. For example, [43] report on the “lack of quality” in automatic text simplifications when generating materials for people with intellectual disabilities, which in turn can affect the results of the study. We argue that standard metrics of complexity at the lexical, syntactic, and discourse levels provide important insights into the quality of LLM-generated text when evaluated with respect to the human baseline.

2.1 Shallow Features: Average Word Length and Sentence Length

Word length and sentence length are the backbones of the standard readability measures, such as the Flesch-Kincaid formula [28], Gunning-Fox index [19], and SMOG [31]. Texts with shorter words and sentences are easier to read and process. However, such measures are rather crude proxies of text complexity. Word length would not necessarily be a deterrent for children if the word refers to familiar entities, such as dinosaur names or names of favorite movie characters [42]. Moreover, studies show that shallow features are not particularly strong predictors of readability judgments [37]. Therefore, a comprehensive analysis of text complexity should also include lexical, syntactic, and discourse features.

2.2 Lexical Features

Lexical simplification pertains to the process of replacing difficult words with the alternatives that are easier to read and understand [5, 36, 37]. Lexical features for measuring text complexity are based on research in psycholinguistics. The MRC Psycholinguistic Database [51] provides word ratings that are implemented in evaluating lexical complexity. The relevant features include age of acquisition, imageability, and concreteness. Age of acquisition pertains to the age at which words are learned. This measure is a good predictor of the ease and speed with which words are recognized during reading [26]. Words that are acquired earlier are recognized and processed faster than words that are acquired later [13, 17]. Age of acquisition is often used to measure lexical complexity in the literature on text simplification for children [39, 48]. These studies show that words in LLM-generated output are more difficult than expected (cf. also [12] on earlier models).

Imageability pertains to a “word’s capacity to arouse nonverbal images” [38]. The word *piano* will score high on the imageability scale, while the word *fact* will score low. As text gets simplified, we expect to see more words that refer to tangible, easily imaginable things [10]. Concreteness refers to “directness of reference to sense experience” [38]. While imageability and concreteness are related, they are not identical. The word *prosperity* will score relatively high on the imageability scale but rather low on the scale of concreteness. Words that score high on imageability and on concreteness scales are easier to understand.

In addition to psycholinguistic measures, lexical complexity can be operationalized in terms of lexical richness (diversity), as a type-to-token ratio (TTR). The more unique words there are, the greater lexical richness of the text. Lexically rich texts are more difficult to read as they assume familiarity with a greater range of vocabulary.

A reviewer suggested that WordNet [32] can also be used to measure semantic complexity. In the previous work, WordNet was used in the context of building text simplification models (cf. [12]; [42] for an overview). In such models, difficult words were replaced with the corresponding synonyms from the WordNet synonym sets (synsets). Since we are not building a text simplification model, WordNet has a more limited applicability in the context of the current work. One possible application is to use semantic information in WordNet as operationalization of abstraction (but see [25]). In what follows, we use the concreteness feature from the MRC database and leave the question of whether WordNet can be leveraged to measure abstraction in conjunction with the MRC database for future work.

2.3 Syntactic Features

There are several measures of syntactic complexity. Dependent clauses, such as subordinate clauses (*Cleo thinks that it will rain*) and relative clauses (*the boy who lives next door*), make sentences syntactically complex. During the process of text simplification such sentences are split into shorter independent sentences, often resulting in a greater length of the simplified texts compared to the original. Additionally, [14] show that complex nominals, including heavily modified nouns (*a large, prowling ginger cat*) and nominalizations, nouns derived from verbs (*their music practice, dreadful meowing*), present additional processing challenge for children (see also [4]). Prepositional clauses can also be used to measure text complexity, as they contribute to ambiguity and thus can increase processing cost [6, 30]. For example, in the sentence *The girl praised the boy with the smile*, the prepositional phrase *with the smile* can refer to either boy or girl. Syntactic simplifications were shown to be effective in improving readability and comprehension for children [41] and should be included in the evaluative metrics to assess LLMs' output.

2.4 Discourse Features

Discourse cohesion measures how well sentences are connected. Referential cohesion can be established by the repeated usage of the same lexical item and can be computed based on the word overlap in text [21]. Global text cohesion is established by the repetition of words throughout the entire passage. Local cohesion is measured based on the overlap of words in the neighboring sentences. In the following example from Ernest Hemingway's short story [23], discourse cohesion at a local level is established via the repeated usage of the word *river* (from [20]).

"Nick looked at the burned-over stretch of hillside, where he had expected to find the scattered houses of the town and then walked down the railroad track to the bridge over the river. The river was there."
(Big Two-Hearted River, Part I by Ernest Hemingway)

Text cohesion was shown to be a good predictor of text quality [40] and is often incorporated into the assessment of text simplification [9].

3 Study

In this study we evaluate GPT-4o's ability to simplify texts for elementary school children. The goal of the study is twofold. First, we establish whether GPT-4o is able to perform simplification tasks by comparing complexity of original texts to that in GPT-4o-simplified texts. Second, we evaluate GPT-4o's performance with respect to the human baseline.

3.1 Materials

The original (non-simplified) texts and human-simplified texts were selected from the Newsela corpus [53]. The corpus consists of 1130 texts produced by professional writers in the Newsela company, a provider of educational materials for K-12 curriculum. According to [2], 90% of the US schools, including 1.5 million teachers and 20 million students, use Newsela. The Newsela corpus consists of parallel texts that differ in their levels of difficulty, from the most complex texts for grade 12 to the least complex texts for grade 2. The Newsela corpus is commonly used to evaluate performance of language technology on text simplification tasks, and we use it here both as a human baseline (simplified text) and as input to GPT-4o (original text).

3.2 Design and Procedure

We constructed three parallel corpora: (i) original (non-simplified) texts from the Newsela corpus; (ii) the corresponding human-simplified texts from Newsela; and (iii) LLM-simplified texts generated with OpenAI API (GPT-4o model). All three corpora were submitted to the Content Text Analysis Platform (CTAP), a web tool that supports automatic text analysis [7]. Each text was analyzed on 15 features that operationalize complexity at shallow textual level, lexical, syntactic, and discourse levels. We describe procedural steps and implementations below.

3.2.1 Text selection from Newsela: original and human-simplified texts. Each original text in Newsela is simplified four times for different reading levels. Level 4 texts are the most simplified texts. Our computational analysis of the Newsela corpus revealed that the largest text sets within Newsela are original (non-simplified) texts for grade 12, and texts simplified for grade 4. To construct our corpora, we randomly selected 100 texts from grade 12 texts (the hardest texts) and matched them with the corresponding 100 texts for grade 4 (the easiest texts). The list of texts selected for the analysis is shown in Supplementary Materials.

3.2.2 Construction of LLM-simplified text corpus. The corpus of LLM-simplified texts was generated using the OpenAI API. 100 texts from the original (non-simplified) text corpus were submitted to the OpenAI API with the instructions to simplify. We choose to collect data via the API rather than the chat interface because it allows us to specify model parameters: model = GPT-4o (the most advanced model at the time of writing), temperature = 1. The temperature parameter controls how random the output is. The

lower the temperature setting, the more deterministic the output. The range of the temperature setting is between 0 and 2.

We used a zero-shot strategy, i.e. no examples of text simplifications were submitted as part of the prompt. Our prompt was formulated as follows: “In approximately 680 words, simplify the text below for a fourth-grade reading level written in the newspaper genre”. The prompt was designed with the following considerations in mind. First, we specified the grade level, so that it matches the grade level of the human-simplified texts. Second, we specified the target length for the simplified articles using the average number of words in human-simplified articles for the same grade level as a reference. Finally, since Newsela materials are in the news genre [53], we specified the genre requirement as part of the prompt. Our early experiments with the prompt showed that when information about genre/style is unspecified, GPT tends to produce texts in more colloquial style, as shown in the example below.

Original non-simplified (Newsela): After years of decline, the piano industry might be on the upswing.

Human-simplified (Newsela): People are buying pianos again. Sales had gone down for years, but they are coming back.

GPT-simplified: Good news, kids! Pianos are making a comeback, and it's music to our ears!

GPT-4o's responses were recorded as 100 individual text files. We examined the files for content and established that there were no blank responses. There were also no request denials, and we received simplifications for all 100 input files. These texts constitute our LLM-simplified corpus.

3.2.3 Common Text Analysis Platform (CTAP). We uploaded the three text corpora, original (non-simplified) texts, human-simplified texts, and LLM-simplified texts to CTAP (<https://sifnos.iwm-tuebingen.de/ctap/>). This platform supports the analysis of large quantities of text and offers a great range of analytical features. Based on the previous literature on the topic [4, 39], we selected 15 features for analysis: (i) shallow features: mean word length and sentence length; (ii) lexical: age or acquisition, imageability, concreteness, and lexical richness; (iii) syntactic: relative clauses, subordinate clauses, dependent clauses, complex nominals, and prepositional phrases; (iv) discourse: local and global cohesion. The full list of CTAP features, their definitions and association with complexity, as well as Feature ID numbers in the CTAP platform, are presented in Appendix A.1. CTAP generated output files with the numeric information about complexity feature values for each text, which we used as input for statistical analyses.

3.3 Results

The statistical analyses and visualizations were generated using Pandas, SciPy, NLTK, and Matplotlib.

3.3.1 Original vs Human-simplified texts. The statistical analyses of complexity features in original (non-simplified) and human-simplified texts showed highly significant differences on all features except for local discourse cohesion. The majority of changes are

in the predicted direction toward greater simplicity, and the differences are highly significant on a paired t-test ($p < 0.001$) (See Appendix A.2). These results align with the analyses reported in the previous literature comparing original and simplified texts in Newsela [53]. These analyses do not bear on our main research question but are used to validate our methodology.

3.3.2 Original vs LLM-simplified texts. The statistical analyses of complexity features reveal that there are significant differences between original (non-simplified) and LLM-simplified texts on all features except for local discourse cohesion. The majority of the observed changes are in the predicted direction toward lesser complexity, but see the discussion about the number of sentences below. The differences are highly significant on a paired t-test ($p < 0.001$). (See Appendix A.3).

3.3.3 Human-simplified vs LLM-simplified texts. The results of the statistical analyses show that LLM-simplified texts and human-simplified texts differ in terms of their complexity. The most robust differences are in the domain of syntax: LLM-simplified texts show a greater syntactic complexity as measured by complex nominals, dependent clauses, prepositional phrases, relative clauses and the overall sentence complexity ratio. We argue below that greater syntactic complexity is also responsible for longer sentences (mean sentence length) and the smaller number of sentences in LLM-simplified texts compared to human-simplified texts. For lexical features we find differences on concreteness and Lexical TTR, but no difference on age of acquisition and imageability. There were no differences at the level of discourse. The results are shown in Table 1.

Figure 1 shows the differences between original (non-simplified) texts, human-simplified texts, and LLM-simplified texts.

3.4 Discussion

Our analyses demonstrate that LLM-simplified texts show reduced complexity compared to the original texts. This suggests that LLMs, and GPT-4o, specifically, can perform text simplification tasks reasonably well without any fine-tuning or few-shot prompting. Moreover, while previous research on LLMs' text simplification for children focuses mostly on lexical features and readability scores [33, 48], we show that GPT-4o simplifies texts at the syntactic level and to some extent at the level of discourse. More importantly, when evaluating GPT-4o-simplified texts with respect to human-simplified texts, we found robust differences in the domain of syntax. GPT-4o-generated texts are syntactically more complex compared to human-generated texts on all features. These differences become apparent in side-by-side comparison of texts in Table 2.

Table 2 shows that human writers simplify text by splitting original complex constructions into independent, stand-alone sentences. Only two sentences in the human-simplified passage are syntactically complex (underlined). On the other hand, GPT-4o-simplified texts retain a higher level of syntactic complexity. The opening sentence has two subordinate clauses. These different syntactic strategies result in the differences in the overall number of sentences: 6 in the human-simplified passage vs. only 2 in the GPT-4o-simplified passage. Moreover, GPT-4o-generated sentences are longer. When

Table 1: Comparative Analysis: Human-simplified vs LLM-simplified (Non-simplified values provided for reference only)

Feature Name	Non-simplified	Human-simplified	LLM-simplified	Human-LLM
Mean Sentence Length in Tokens	M=21.86, SD=3.2	M=11.18, SD=0.84	M=15.95, SD=1.9	p<0.001, t=23.57
Mean Token Length in Letters	M=4.81, SD=0.22	M=4.47, SD=0.19	M=4.5, SD=0.21	p=0.07, t=1.84
Number of Sentences	M=45.18, SD=16.06	M=53.32, SD=7.56	M=33.17, SD=7.39	p<0.001, t=24.73
Lexical Richness: TTR	M=13.74, SD=1.78	M=10.5, SD=0.98	M=10.62, SD=0.82	p=0.23, t=1.22
Lexical: Age of Acquisition	M=368.61, SD=23	M=330.82, SD=23.24	M=332.66, SD=28.42	p=0.45, t=0.75
Lexical: Concreteness	M=375.18, SD=14.58	M=361.31, SD=13.94	M=356.03, SD=13.64	p<0.001, t=3.72
Lexical: Imageability	M=405.14, SD=13.24	M=393.54, SD=13.38	M=391.43, SD=13.28	p=0.1, t=1.64
Lexical: Variation (Lexical TTR)	M=0.61, SD=0.05	M=0.55, SD=0.04	M=0.59, SD=0.04	p<0.001, t=9.66
Syntactic: Complex Nominals per Sentence	M=2.94, SD=0.58	M=1.25, SD=0.19	M=1.86, SD=0.4	p<0.001, t=17.62
Syntactic: Dependent clauses per Sentence	M=0.94, SD=0.23	M=0.45, SD=0.12	M=0.77, SD=0.24	p<0.001, t=13.54
Syntactic: Prepositional Phrases per Sentence	M=1.85, SD=0.36	M=0.84, SD=0.15	M=1.28, SD=0.25	p<0.001, t=17.22
Syntactic: Relative Clauses per Sentence	M=0.35, SD=0.13	M=0.13, SD=0.06	M=0.22, SD=0.11	p<0.001, t=7.38
Syntactic: Sentence Complexity Ratio	M=2.07, SD=0.28	M=1.45, SD=0.14	M=1.85, SD=0.25	p<0.001, t=15.91
Discourse: Mean Global Lexical Overlap	M=0.01, SD=0	M=0.02, SD=0.01	M=0.02, SD=0.01	p=0.31, t=1.02
Discourse: Mean Local Noun Overlap	M=0.01, SD=0.01	M=0.01, SD=0.01	M=0.01, SD=0.01	p=0.67, t=0.43

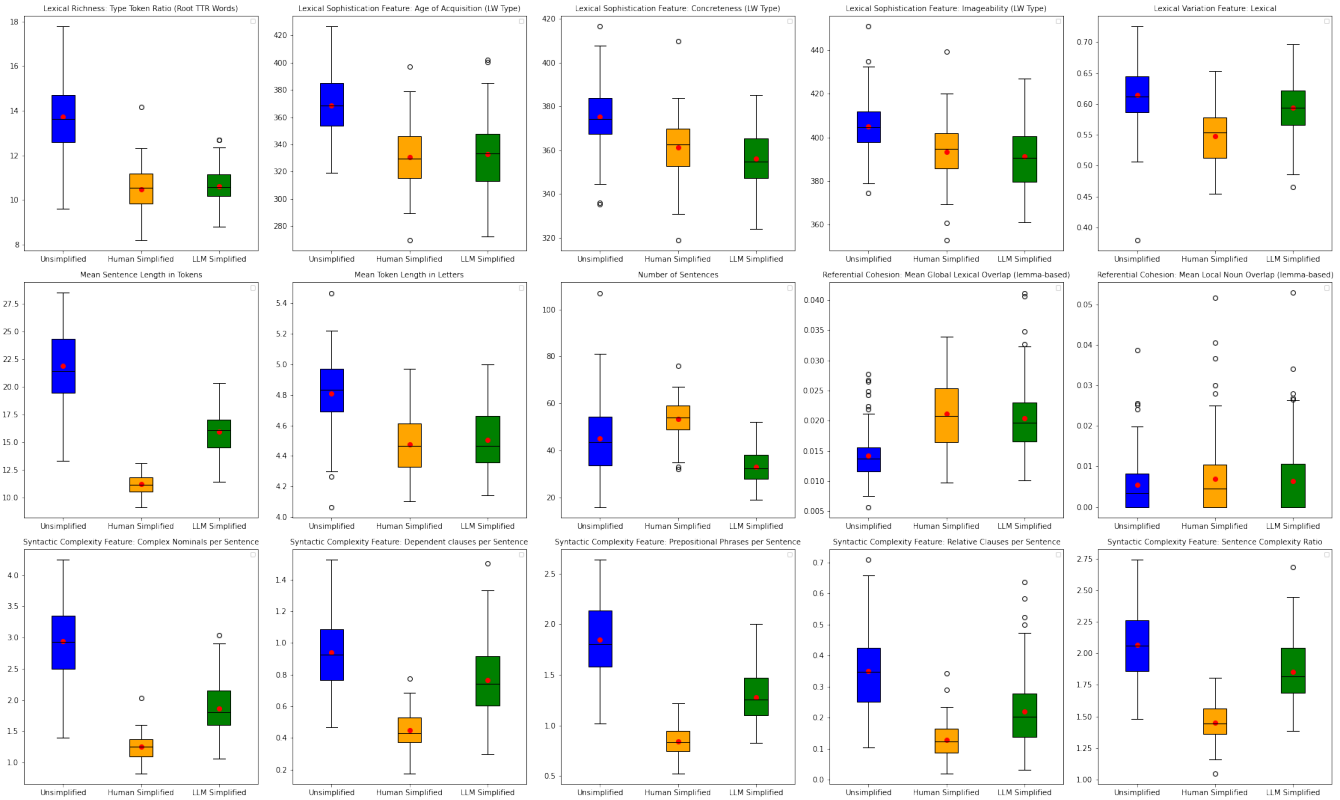
**Figure 1: The differences between non-simplified, human-simplified and LLM-simplified texts on the 15 complexity features. The blue box represents original texts, yellow box – human-simplified, and green box– LLM-simplified texts. The red dot represents the means.**

Table 2: Comparative Analysis: Syntax of Non-simplified, Human-simplified, and LLM-simplified Texts

Text Type	Example
Original (non-simplified)	In the summer of 1914, with the war in Europe just two weeks old, Henry James knew that something had been lost forever. "Black and hideous to me is the tragedy that gathers, and I'm sick beyond cure to have lived on to see it," the American author, an expatriate in London at the time, wrote to a friend.
Human-simplified	World War I began a hundred years ago this month. It was a long war that changed the way people saw the world. One of these people was American writer Henry James. In the summer of 1914, James knew that something had been lost forever. World War I was just two weeks old at the time. James was living in London.
LLM-simplified	In summer 1914, <u>when a big war started in Europe</u> , many people felt <u>things would never be the same again</u> . Henry James, a writer then living in London, felt very sad about the war.

taken together, a greater number of syntactically complex constructions and fewer but longer sentences in LLM output are indicative of tight information packaging or more compressed prose, which is usually associated with higher processing cost.

Since the main focus of this study is how text simplification is manifested at lexical, syntactic, and discourse levels, we did not explicitly analyze how well GPT-4o conveys the meaning of the original text. We manually examined ten texts for hallucinations and content and did not find any apparent mismatches. The most obvious difference that we noticed was that GPT-4o tended to generate headlines and section titles, even if the original texts did not have any. These headlines accurately reflected the meaning of the texts. One possible downside is that that news headlines generated by GPT-4o were written in a highly compressed style (cf. LONDON – New School Book Rules Spark Debate). Reading and understanding headlines can present an additional processing challenge for children.

4 Conclusion

We report two main findings. First, when instructed to simplify text for the specific grade level, GPT-4o, the latest and most advanced LLM at the time of writing, can successfully reduce complexity at lexical, syntactic, and discourse level. This emergent ability can potentially be used to further personalize LLMs' output to address individual needs of different users [18]. Second, when evaluating GPT-4o with respect to the human baseline, we observed that its simplifications are syntactically more complex compared to the human-generated simplifications. This finding is important since children were previously shown to benefit from syntactically simplified texts [41]. Our results suggest the need to further reduce syntactic complexity of LLM-generated texts before it reaches the target audience.

Acknowledgments

We are grateful to the audience at the UC Davis Symposium on Language Research for comments and feedback.

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A Appendix

A.1 Text Complexity Features from CTAP

Table 3 provides detailed descriptions of the CTAP features used for the analysis of text complexity.

A.2 Analysis of Non-simplified vs Human-simplified Texts from Newsela

Table 4 provides results of the statistical analyses comparing non-simplified and human-simplified articles from Newsela.

Table 3: CTAP Features for Measuring Text Complexity

Feature Name in CTAP	Feature ID	Type	Description (CTAP)	As text gets simpler, will ...
Mean Sentence Length in Tokens	719	shallow	Calculates the mean sentence length in number of tokens.	decrease
Mean Token Length in Letters	177	shallow	Calculates the mean token length in letters.	decrease
Number of Sentences	343	shallow	Calculates the number of sentences in a text.	increase
Lexical Richness: Type Token Ratio (Root TTR Words)	382	lexical	Calculates TTR for words (excl. punctuation and numbers) $\text{LogTTR} = T/\sqrt{N}$.	decrease
Lexical Sophistication Feature: Age of Acquisition (LW Type)	332	lexical	Calculates age of acquisition of lexical words (LW) from a norm list by [17], included in the MRC Psycholinguistic Database. Lexical words (LW): verbs, nouns, adverbs and adjectives.	decrease
Lexical Sophistication Feature: Concreteness (LW Type)	781	lexical	Calculates concreteness from the norm list in the MRC Psycholinguistic Database.	increase
Lexical Sophistication Feature: Imageability (LW Type)	814	lexical	Calculates imageability from the norm list in the MRC Psycholinguistic Database.	increase
Lexical Variation Feature: Lexical	1099	lexical	Calculates Lexical Type/Token ratio. Lexical words are verbs, nouns, adverbs and adjectives.	decrease
Syntactic Complexity Feature: Complex Nominals per Sentence	918	syntactic	Calculates the complex nominals per sentence	decrease
Syntactic Complexity Feature: Dependent clauses per Sentence	810	syntactic	Calculates the dependent clause per sentence.	decrease
Syntactic Complexity Feature: Prepositional Phrases per Sentence	449	syntactic	Calculates the prepositional phrases per sentence.	decrease
Syntactic Complexity Feature: Relative Clauses per Sentence	642	syntactic	Calculates the relative clauses per sentence.	decrease
Syntactic Complexity Feature: Sentence Complexity Ratio	1008	syntactic	Calculates the sentence complexity ratio by dividing the number of clauses by the number of sentences.	decrease
Referential Cohesion: Mean Global Lexical Overlap (lemma-based)	848	discourse	Calculates the referential cohesion of the text. This measure calculates the mean global lexical overlap based on lemmas.	increase
Referential Cohesion: Mean Local Noun Overlap (lemma-based)	835	discourse	Calculates the mean local noun lemma overlap.	increase

Table 4: Comparative Analysis: Non-simplified vs Human-simplified

Feature Name	Non-simplified	Human-simplified	Non-simplified vs Human-simplified paired t-test
Mean Sentence Length in Tokens	M=21.86, SD=3.2	M=11.18, SD=0.84	p<0.001, t=33.2
Mean Token Length in Letters	M=4.81, SD=0.22	M=4.47, SD=0.19	p<0.001, t=18.55
Number of Sentences	M=45.18, SD=16.06	M=53.32, SD=7.56	p<0.001, t=5.49
Lexical Richness: TTR	M=13.74, SD=1.78	M=10.5, SD=0.98	p<0.001, t=21.43
Lexical: Age of Acquisition	M=368.61, SD=23	M=330.82, SD=23.24	p<0.001, t=17.23
Lexical: Concreteness	M=375.18, SD=14.58	M=361.31, SD=13.94	p<0.001, t=9.94
Lexical: Imageability	M=405.14, SD=13.24	M=393.54, SD=13.38	p<0.001, t=9.13
Lexical: Variation (Lexical TTR)	M=0.61, SD=0.05	M=0.55, SD=0.04	p<0.001, t=12.84
Syntactic: Complex Nominals per Sentence	M=2.94, SD=0.58	M=1.25, SD=0.19	p<0.001, t=30.85
Syntactic: Dependent clauses per Sentence	M=0.94, SD=0.23	M=0.45, SD=0.12	p<0.001, t=23.1
Syntactic: Prepositional Phrases per Sentence	M=1.85, SD=0.36	M=0.84, SD=0.15	p<0.001, t=28.68
Syntactic: Relative Clauses per Sentence	M=0.35, SD=0.13	M=0.13, SD=0.06	p<0.001, t=17.64
Syntactic: Sentence Complexity Ratio	M=2.07, SD=0.28	M=1.45, SD=0.14	p<0.001, t=24.01
Discourse: Mean Global Lexical Overlap	M=0.01, SD=0	M=0.02, SD=0.01	p<0.001, t=11.31
Discourse: Mean Local Noun Overlap	M=0.01, SD=0.01	M=0.01, SD=0.01	p=0.21, t=1.25

A.3 Analysis of Non-simplified vs LLM-simplified Texts from Newsela

Table 5 provides results of the statistical analyses comparing non-simplified and LLM-simplified articles from Newsela.

Table 5: Comparative Analysis: Non-simplified vs LLM-simplified

Feature Name	Non-simplified	LLM-simplified	Non-simplified vs LLM-simplified paired t-test
Mean Sentence Length in Tokens	M=21.86, SD=3.2	M=15.95, SD=1.9	p<0.001, t=19.47
Mean Token Length in Letters	M=4.81, SD=0.22	M=4.5, SD=0.21	p<0.001, t=13.65
Number of Sentences	M=45.18, SD=16.06	M=33.17, SD=7.39	p<0.001, t=8.83
Lexical Richness: TTR	M=13.74, SD=1.78	M=10.62, SD=0.82	p<0.001, t=19.08
Lexical: Age of Acquisition	M=368.61, SD=23	M=332.66, SD=28.42	p<0.001, t=16.61
Lexical: Concreteness	M=375.18, SD=14.58	M=356.03, SD=13.64	p<0.001, t=15.39
Lexical: Imageability	M=405.14, SD=13.24	M=391.43, SD=13.28	p<0.001, t=12.02
Lexical: Variation (Lexical TTR)	M=0.61, SD=0.05	M=0.59, SD=0.04	p<0.001, t=4.27
Syntactic: Complex Nominals per Sentence	M=2.94, SD=0.58	M=1.86, SD=0.4	p<0.001, t=20.25
Syntactic: Dependent clauses per Sentence	M=0.94, SD=0.23	M=0.77, SD=0.24	p<0.001, t=7.37
Syntactic: Prepositional Phrases per Sentence	M=1.85, SD=0.36	M=1.28, SD=0.25	p<0.001, t=16.66
Syntactic: Relative Clauses per Sentence	M=0.35, SD=0.13	M=0.22, SD=0.11	p<0.001, t=9.05
Syntactic: Sentence Complexity Ratio	M=2.07, SD=0.28	M=1.85, SD=0.25	p<0.001, t=7.37
Discourse: Mean Global Lexical Overlap	M=0.01, SD=0	M=0.02, SD=0.01	p<0.001, t=10.22
Discourse: Mean Local Noun Overlap	M=0.01, SD=0.01	M=0.01, SD=0.01	p=0.36, t=0.91